

1 SQL TRANSACTION

Why we use Transactions

We use transactions to **maintain data accuracy and safety** when multiple SQL statements must succeed together.

👉 If one statement fails, **all changes are undone**.

Importance

- Prevents **partial updates**
- Protects **data integrity**
- Essential for **financial & critical systems**

Advantages of Transactions

- ✓ Ensures **data consistency**
- ✓ Avoids **data corruption**
- ✓ Supports **rollback** if error occurs
- ✓ Controls **concurrent access**
- ✓ Reliable for banking, billing, orders

Disadvantages of Transactions

- ✗ Can **reduce performance**
- ✗ Locks tables/rows (may cause delays)
- ✗ Long transactions can cause **deadlocks**
- ✗ Requires careful error handling

Example

sql

```
BEGIN TRANSACTION;  
UPDATE orders SET status = 'Paid' WHERE order_id = 10;  
UPDATE inventory SET quantity = quantity - 1 WHERE product_id = 5;  
COMMIT;
```

2 STORED PROCEDURE

Why we use Stored Procedures

We use stored procedures to **store SQL logic inside the database** and execute it whenever needed.

Importance

- Improves **code reusability**
- Enhances **security**
- Increases **performance**

Advantages of Stored Procedures

- ✓ Reusable code
- ✓ Faster execution (precompiled)
- ✓ Reduces network traffic
- ✓ Improves security (no direct table access)
- ✓ Easier maintenance

Disadvantages of Stored Procedures

- ✗ Harder to debug
- ✗ Database-dependent (not portable)
- ✗ Requires skilled developers
- ✗ Can increase DB server load

Example

```
sql

CREATE PROCEDURE GetStudent
    @Id INT
AS
BEGIN
    SELECT * FROM Students WHERE StudentID = @Id;
END;
```

3 COMBINED IMPORTANCE

(Transaction + Stored Procedure)

Stored procedures often **use transactions internally** to ensure safe execution.

Example Use Case

- ✓ Bank transfer
- ✓ Online shopping checkout
- ✓ Salary processing

Summary Table (Easy to Remember)

Feature	Transaction	Stored Procedure
Purpose	Data safety	Code reusability
Rollback	Yes	No (but can use transaction)
Performance	Slower	Faster
Security	Medium	High

★ Exam Tip

Transaction = Safety
Stored Procedure = Performance + Security